Ivan Sazontov

Personal details

Ivan Sazontov

isazontov1@gmail.com

Ho Chi Minh City, Vietnam - Open to remote

t.me/Hammerschmidt1

github.com/greencat1/Portfolio

Skills

Deep Learning / Computer Vision:
- PyTorch
- TensorFlow/Keras
- Ultralytics
- YOLO
- CNN
- Transfer learning

Image Processing & Augmentation:
- OpenCV
- Albumentations

Data Analysis & Classical ML:
- Pandas
- NumPy
- Matplotlib
- Seaborn
- scikit-learn
- CatBoost
- SciPy
- EDA (Exploratory Data Analysis)
- Feature engineering
- Hyperparameter tuning
- Cross-validation
- Class imbalance handling

Vector Search & Embeddings (RAG):
- Vector Databases
- FAISS
- Sentence-Transformers

Local LLMs & Generative AI:
- Ollama

Profile

Recent Master's graduate in Applied Mathematics with practical experience in developing machine learning solutions through freelance projects. Built several end-to-end ML systems including Computer Vision, predictive analytics, and RAG applications. Focused on model training, deployment (FastAPI + Docker) and achieving strong performance metrics. Actively seeking remote Machine Learning Engineer or Data Scientist positions.

Education

Master’s Degree in Applied Mathematics — **2024**
Moscow Aviation Institute (National Research University)
- with honors

Bachelor’s Degree in Applied Mathematics — **2022**
Moscow Aviation Institute (National Research University)

Employment

Freelance Data Scientist — **2022 - Present**

- Delivered end-to-end ML solutions for clients in computer vision, NLP, time series forecasting, and game theory.
- Assisted clients with model implementation, data processing, experimentation, and technical documentation for research projects.
- Designed, trained, and deployed models, including production-ready web services using FastAPI and Docker.

Courses

Applied Data Science with Python Specialization

More Applied Data Science with Python Specialization

Data Science Fundamentals with Python and SQL

Key Projects

Two-Stage Traffic Sign Detection System (YOLO + ResNet)
Real-time recognition of 55 traffic sign classes
- Built hybrid cascade architecture: YOLO for coarse detection (5 classes) + 5 specialized ResNet-18 models for fine classification.
- Achieved 94.07% mAP50 (+12.7% over single-stage baseline) and 91.71% Recall.
- Implemented real-time inference via WebSocket with browser camera

- OpenAI
- Prompt Engineering

Backend & Deployment:
- FastAPI
- Docker
- Docker Compose
- Authentication (API keys, RBAC)

ML Prototyping & UI:
- Streamlit
- PyQT

Testing:
- pytest

Data Handling & SQL:
- SQL
- PostgreSQL (basics)
- SQLite

Development Tools & OS:
- Git | GitHub
- Linux / bash
- VPS
- AWS
- Kubernetes
- CI/CD
- MLflow

Statistics & Probability:
- Descriptive statistics
- Hypothesis testing (t-test, chi-square)
- Distributions (Normal, Poisson, Binomial
- Central Limit Theorem (CLT)
- Bootstrapping
- A/B testing

Business Skills:
- ROI calculation
- Economic impact (MRR)
- Technical documentation

Languages

Russian - Native

English - B2

German - B2

support.
- Stack: PyTorch, YOLO, ResNet-18, FastAPI, Docker, Albumentations
- Live Demo: https://greencat1.tech

Customer Churn Prediction Service

End-to-end MLOps solution for telecom
- Developed a full production service with 23 REST API endpoints, automated retraining pipeline, manual labeling, and role-based access.
- Achieved ~96% Recall on churn prediction using CatBoost model.
- Built interactive Streamlit dashboard for churn analysis and business insights.
- Stack: FastAPI, CatBoost, Streamlit, SQLite, Docker Compose, pytest
- Live Demo Dashboard: http://207.231.105.113:8501 - key - dashboard
- Live Demo Api - http://207.231.105.113:8000/docs# - key admin and user

Local RAG Portfolio Assistant

Fully local Retrieval-Augmented Generation system
- Processed 8,361 semantic chunks from 291 files, including AST parsing of Python code.
- Implemented complete RAG pipeline with vector search and local LLM generation.
- Stack: FastAPI, FAISS, Sentence-Transformers (BGE), Ollama, Streamlit, Docker Compose
- Live Demo: http://207.231.105.113:8502

Publications

Traffic Sign Detection Using SSD **2024**

International Conference on Applied Mathematics, Computer Science and Mechanics, Voronezh, 2024

Sazontov I.Yu.

Application of YOLOv7 for Traffic Sign Detection **2023**

Gagarin Readings, Moscow, 2023

Sazontov I.Yu.